Experiment 8

Aim:

To set up a cryptocurrency wallet using MetaMask and connect it to a decentralized application (DApp) to execute and verify blockchain transactions.

Theory:

MetaMask is a digital cryptocurrency wallet that enables users to access blockchain networks and interact with decentralized applications. It manages blockchain accounts and allows users to approve or reject transactions requested by DApps.

A Decentralized Application (DApp) is an application that interacts with smart contracts deployed on a blockchain. Its frontend connects to the blockchain through a Web3 provider, such as MetaMask, which serves as a link between the user's browser and the blockchain network.

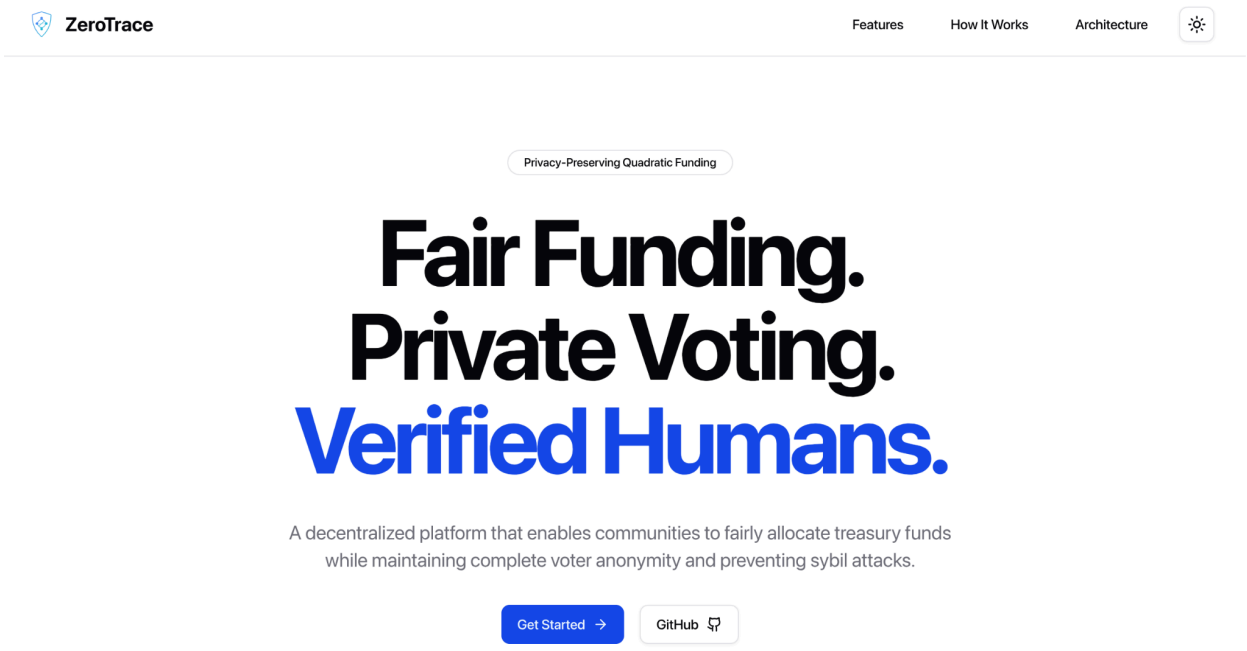
In this experiment, the ZeroTrace DApp was used to demonstrate the integration of MetaMask with a decentralized application. The DApp uses Web3 technologies to connect to the user's wallet and communicate with deployed smart contracts. After connecting the wallet, a blockchain transaction was initiated through the DApp and approved by the user in MetaMask. The transaction was then verified using its unique transaction hash on a blockchain explorer.

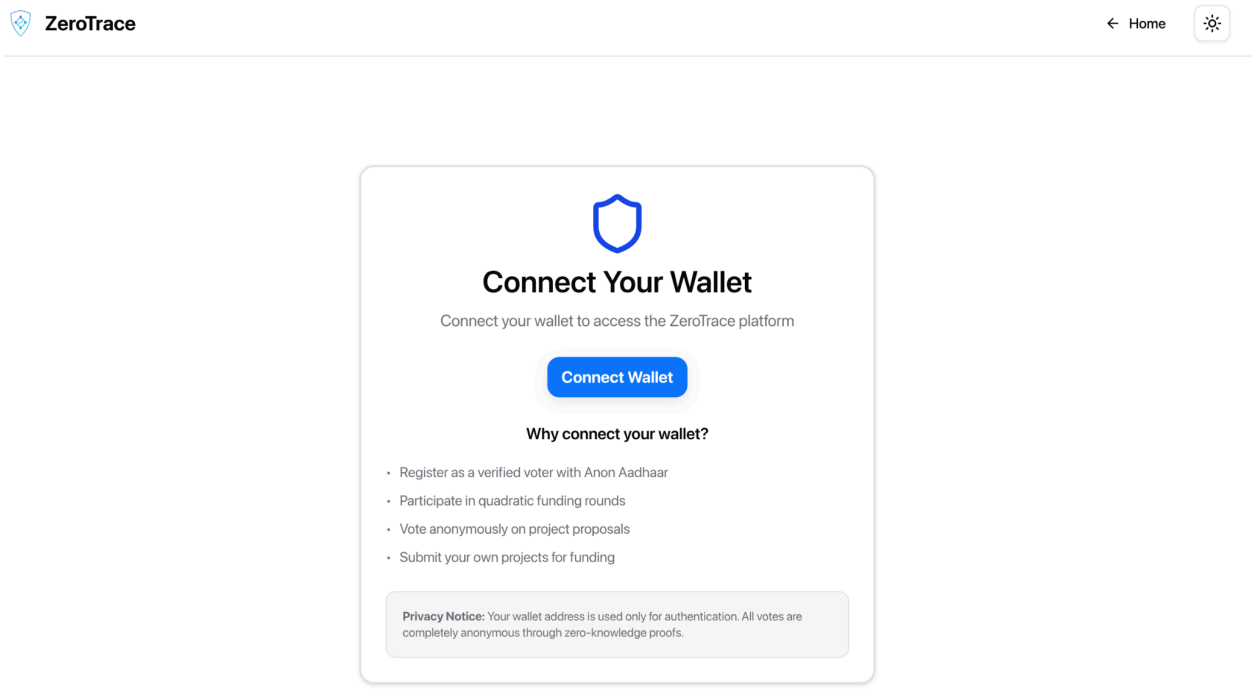
The basic interaction flow is represented as:

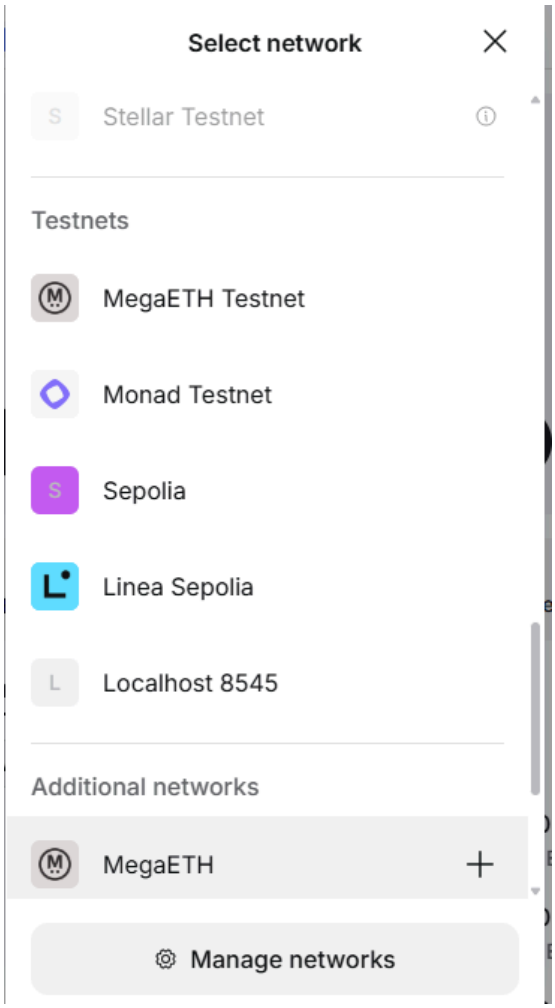
User → ZeroTrace DApp → MetaMask Wallet → Smart Contract → Blockchain Network → Transaction Confirmation

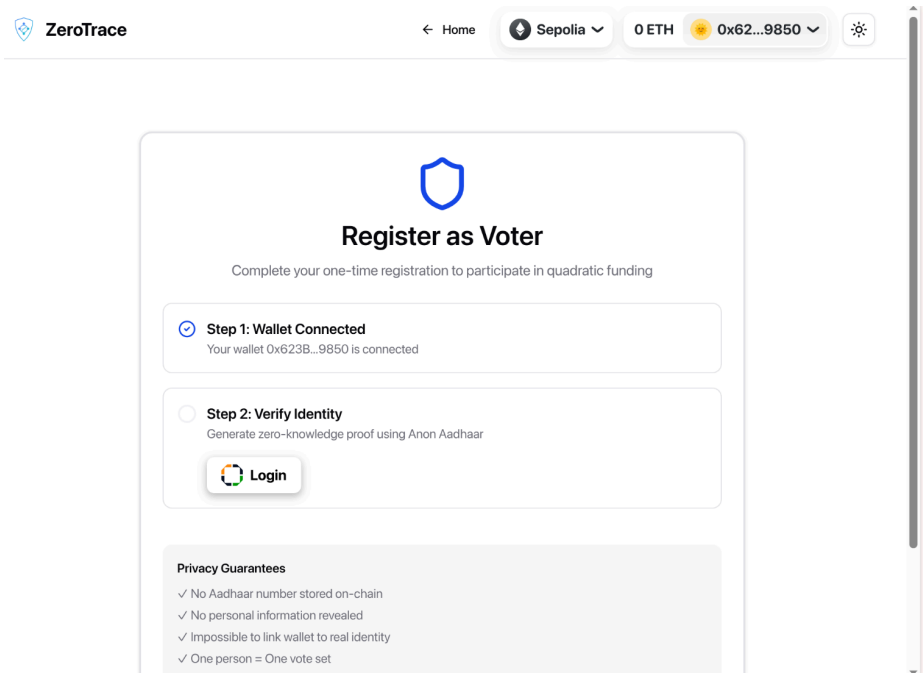
MetaMask provides features such as account management, network selection, transaction authorization, and digital signing. Each submitted blockchain transaction receives a unique transaction hash, which can be used to check its status and view its details on a blockchain explorer.

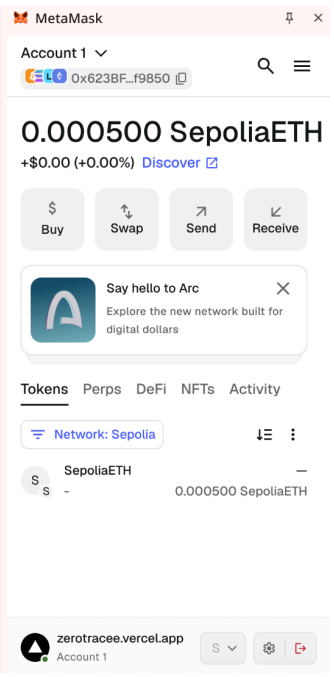
ZeroTrace DApp

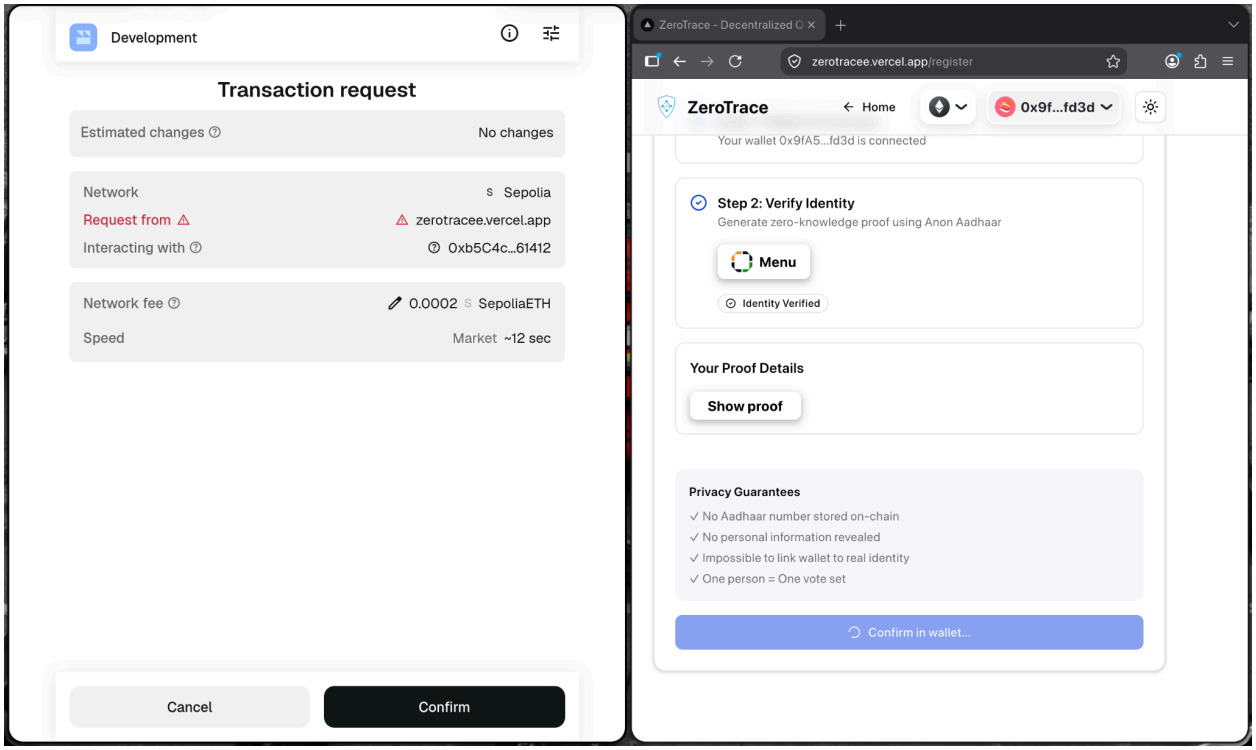

ZeroTrace
Features
How It Works
Architecture
Privacy-Preserving Quadratic Funding
Fair Funding.
Private Voting.
Verified Humans.
A decentralized platform that enables communities to fairly allocate treasury funds while maintaining complete voter anonymity and preventing sybil attacks.
Get Started
GitHub

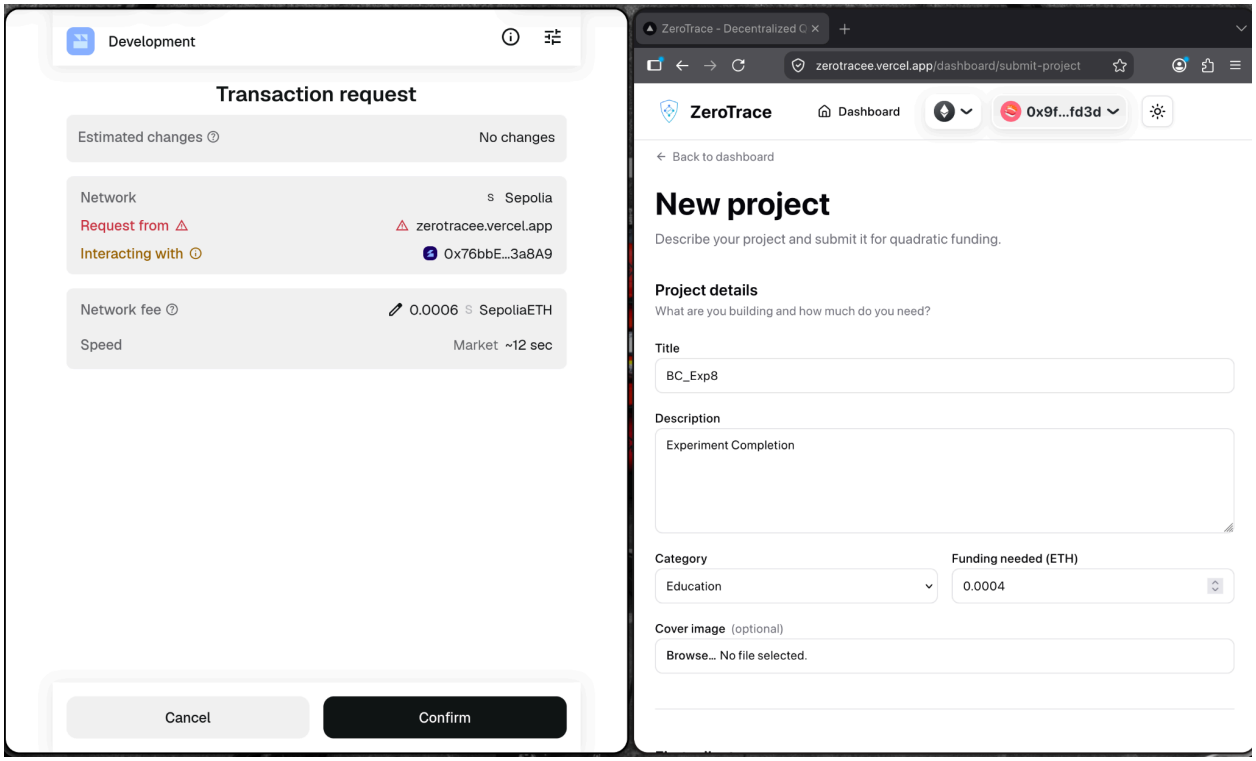

ZeroTrace
Home
Connect Your Wallet
Connect your wallet to access the ZeroTrace platform
Connect Wallet
Why connect your wallet?
Register as a verified voter with Anon Aadhaar
Participate in quadratic funding rounds
Vote anonymously on project proposals
Submit your own projects for funding
Privacy Notice: Your wallet address is used only for authentication. All votes are completely anonymous through zero-knowledge proofs.

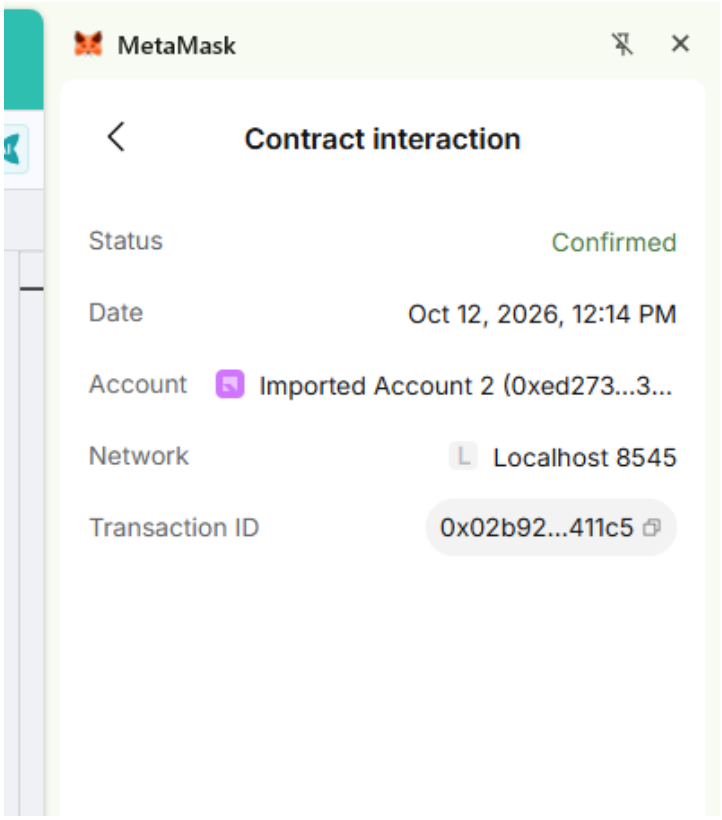

Select network
Stellar Testnet
Testnets
MegaETH Testnet
Monad Testnet
Sepolia
Linea Sepolia
Localhost 8545
Additional networks
MegaETH
Manage networks

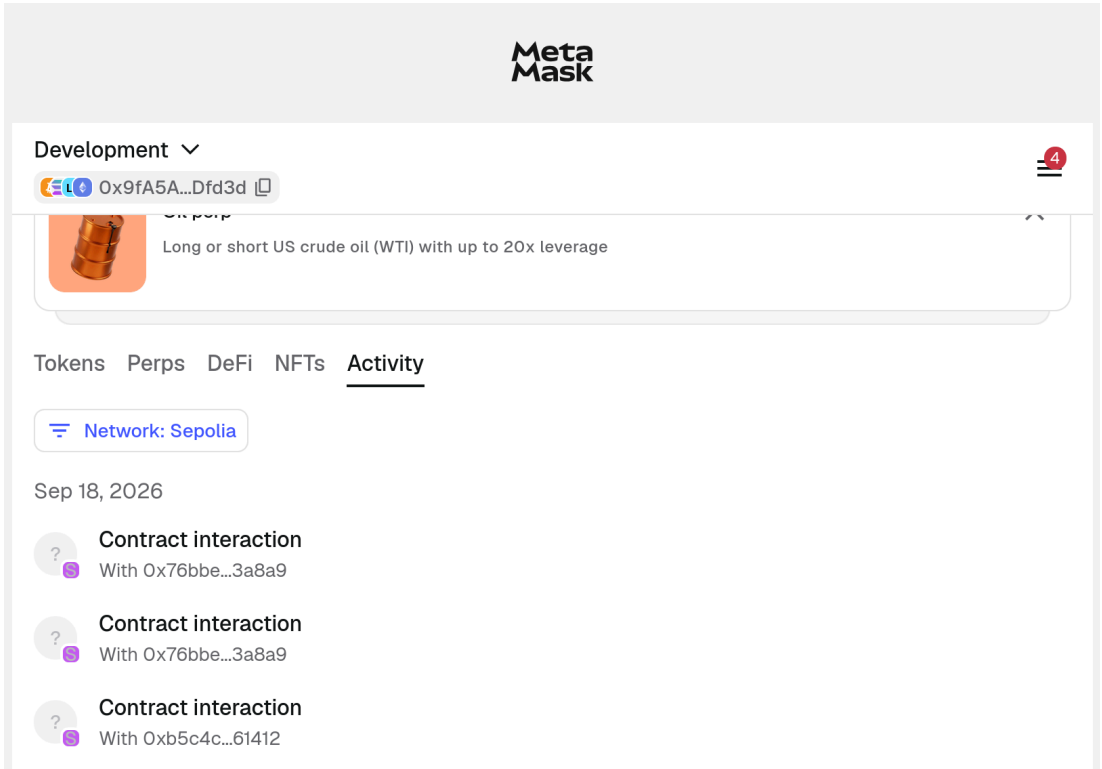

ZeroTrace
Home
Sepolia
0 ETH
0x62...9850
Register as Voter
Complete your one-time registration to participate in quadratic funding
Step 1: Wallet Connected
Your wallet 0x623B...9850 is connected
Step 2: Verify Identity
Generate zero-knowledge proof using Anon Aadhaar
Login
Privacy Guarantees
✓ No Aadhaar number stored on-chain
✓ No personal information revealed
✓ Impossible to link wallet to real identity
✓ One person = One vote set

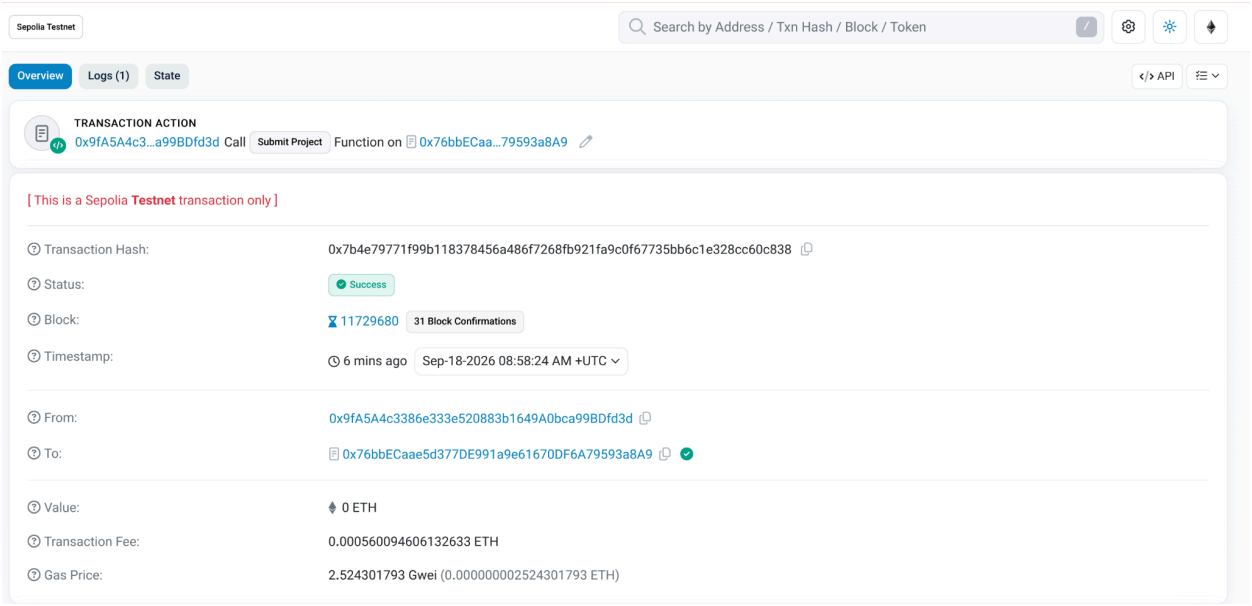

MetaMask
Account 1
0x623BF...f9850
0.000500 SepoliaETH
+$0.00 (+0.00%) Discover
Buy
Swap
Send
Receive
Say hello to Arc
Explore the new network built for digital dollars
Tokens
Perps
DeFi
NFTs
Activity
Network: Sepolia
SepoliaETH
0.000500 SepoliaETH
zerotracee.vercel.app
Account 1


Development
Transaction request
Estimated changes
No changes
Network
Sepolia
Request from
zerotracee.vercel.app
Interacting with
0xb5C4c...61412
Network fee
0.0002 SepoliaETH
Speed
Market ~12 sec
Cancel
Confirm
ZeroTrace - Decentralized
zerotracee.vercel.app/register
ZeroTrace
Home
0x9f...fd3d
Your wallet 0x9fA5...fd3d is connected
Step 2: Verify Identity
Generate zero-knowledge proof using Anon Aadhaar
Menu
Identity Verified
Your Proof Details
Show proof
Privacy Guarantees
✓ No Aadhaar number stored on-chain
✓ No personal information revealed
✓ Impossible to link wallet to real identity
✓ One person = One vote set
Confirm in wallet...


Development
Transaction request
Estimated changes
No changes
Network
Sepolia
Request from
zerotracee.vercel.app
Interacting with
0x76bbE...3a8A9
Network fee
0.0006 SepoliaETH
Speed
Market ~12 sec
Cancel
Confirm
ZeroTrace - Decentralized
zerotracee.vercel.app/dashboard/submit-project
ZeroTrace
Dashboard
0x9f...fd3d
Back to dashboard
New project
Describe your project and submit it for quadratic funding.
Project details
What are you building and how much do you need?
Title
BC_Exp8
Description
Experiment Completion
Category
Education
Funding needed (ETH)
0.0004
Cover image (optional)
Browse... No file selected.


MetaMask
Contract interaction
Status
Confirmed
Date
Oct 12, 2026, 12:14 PM
Account
Imported Account 2 (0xed273...3...
Network
Localhost 8545
Transaction ID
0x02b92...411c5


MetaMask
Development
0x9fA5A...Dfd3d
Long or short US crude oil (WTI) with up to 20x leverage
Tokens
Perps
DeFi
NFTs
Activity
Network: Sepolia
Sep 18, 2026
Contract interaction
With 0x76bbe...3a8a9
Contract interaction
With 0x76bbe...3a8a9
Contract interaction
With 0xb5c4c...61412



Conclusion:

Thus, the MetaMask cryptocurrency wallet was successfully configured and integrated with the ZeroTrace DApp. The DApp was able to connect to the MetaMask wallet and initiate a blockchain transaction, which was approved by the user through MetaMask. The transaction was successfully recorded and verified using its transaction ID on the blockchain explorer. This experiment demonstrated the interaction between a cryptocurrency wallet, a decentralized application, smart contracts, and a blockchain network.